GENERAL INFORMATION:

SIMPLWINDOWS NAME:	Qsys Softphone Controller
CATEGORY:	Qsys
VERSION:	5.0.0
SUMMARY:	Controls a Q-SYS Softphone component: dialing, answering and hanging up calls, DTMF keypad, auto answer, do not disturb and the recent calls list, with call state feedback. The Softphone component is addressed as a Named Component by its Code Name (ComponentName parameter).
GENERAL NOTES:	The dial string is built and held in the Crestron module (shown on DialStringOut). It is only sent to Q-SYS (call_number control) when Dial or Redial is pulsed, which then also triggers call_connect. A full number can be entered on DialString (for example from a touch panel text entry) instead of using the keypad. While a call is connected, keypad presses are also sent to Q-SYS as DTMF (call_pinpad controls). They are still appended to DialStringOut. The dial string is cleared when a call is dialed and when the call disconnects. Connected, Rining, AutoAnswerStatus and DndStatus follow the call_offhook, call_ringing, call_autoanswer and call_dnd controls. Dialing and IncomingCall are derived from the text of the call_status control ("Dialing" and "Incoming Call"). Selecting a recent call loads its number (the text up to the first space) into the dial string; it does not dial. Pulse Dial to place the call. AutoAnswerToggle and DndToggle use the current feedback to decide the new state, so feedback must have been received from Q-SYS for a toggle to act correctly.
CRESTRON HARDWARE REQUIRED:	3 Series Processor or 4 Series Processor
SETUP OF CRESTRON HARDWARE:	Requires a Qsys Core module in the same program with a matching CoreID.
VENDOR FIRMWARE:	N/A
VENDOR SETUP:	In Q-SYS Designer, set the Softphone component's Code Name to match the ComponentName parameter and set its Script Access to External (or All).
GENERATION:	Generated by Claude Code from SIMPL+ and SIMPL# source files

CONTROL:

Dial	D	Pulse to dial the number in the dial string (DialStringOut). Sends the number to Q-SYS and connects.
[Redial]	D	Pulse to redial the last number dialed from this module. The last number is stored when a call dialed with Dial or Redial disconnects.
Connect	D	Pulse to answer an incoming call (pulses call_connect). Does not send the dial string; use Dial to place a call.

Disconnect	D	Pulse to hang up the current call (pulses call_disconnect).
[AutoAnswerToggle]	D	Pulse to toggle auto answer on/off.
[DndToggle]	D	Pulse to toggle do not disturb on/off.
[KeyPadDelete]	D	Pulse to remove the last character from the dial string.
[KeyPadClear]	D	Pulse to clear the dial string.
[KeyPadStar]	D	Pulse to add * to the dial string. Sends DTMF * while a call is connected.
[KeyPadPound]	D	Pulse to add # to the dial string. Sends DTMF # while a call is connected.
KeyPad00–KeyPad09	D	Pulse to add the digit 0–9 to the dial string. Sends the DTMF digit while a call is connected.
[SelectRecentCall[1]]–[SelectRecentCall[5]]	D	Pulse to load the corresponding recent call (RecentCalls[1]–RecentCalls[5]) into the dial string.
[SelectRecentCallIndex]	A	Loads the recent call at this position in the Q-SYS recent calls list (1 = most recent entry) into the dial string. Acts when the value changes; 0 and positions beyond the end of the list are ignored.
[DialString]	S	Appends the received text to the dial string (it does not replace it; pulse KeyPadClear first to start a new number). Ignored while a call is connected. Pulse Dial to place the call.

FEEDBACK:

Connected	D	High while the line is off hook (call connected or in progress).
[Rining]	D	High while the line is ringing.
[Dialing]	D	High while the Q-SYS call status shows the call is dialing.
[IncomingCall]	D	High while the Q-SYS call status shows an incoming call.
[AutoAnswerStatus]	D	High when auto answer is on.
[DndStatus]	D	High when do not disturb is on.
[CurrentlyCalling]	S	Number dialed from this module, shown once the call goes off hook. Cleared when the call disconnects.
[CallStatus]	S	Call status text from the Q-SYS call_status control.
DialStringOut	S	Current dial string being built with the keypad, recent call selection or DialString.
[RecentCallXSig]	S	Intended to output the recent calls list as XSig for a touch panel list. Not currently populated by the module; use RecentCalls[1]–RecentCalls[5].
[RecentCalls[1]]–[RecentCalls[5]]	S	Text of the first five entries in the Q-SYS recent calls list. Unused entries are empty.

PARAMETERS:

CoreID	S	Used to address this module to a Qsys Core module. Must match the CoreID of the Qsys Core module.
ComponentName	S	Code Name of the Softphone component in the Q-SYS design.

TESTING:

SAMPLE PROGRAM:	3-Series: 3 Series Test.smw 4-Series: 4 Series Test.smw
REVISION HISTORY:	